

Modeling occupant behavior's influence on the energy efficiency of solar domestic hot water systems

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HIGHLIGHTS

- An information-driven stochastic domestic hot water usage model was proposed.
- System optimization based on water usage can increase solar energy utilization.
- Improve the location of the reheater can increase solar energy utilization by 30%.
- Solar energy usage may sometimes conflict with energy-saving awareness of occupants.

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ABSTRACT

Solar hot water (SHW) systems are widely used to save energy and reduce CO₂ emissions. The domestic hot water (DHW) usage behavior of occupants has a significant influence on the energy consumption of SHW systems; the operation of SHW systems also affects the DHW usage of occupants. However, there is a lack of quantitative analysis for coordinating DHW behavior and the design and operation of SHW systems to maximize solar energy utilization and reduce energy consumption. Therefore, an information-driven stochastic DHW usage model that interacts with SHW systems is proposed. The optimal design and control strategies for SHW systems to match user behaviors were explored. Meanwhile, the effects of occupants' sensitivity to water temperature and their energy-saving awareness on the energy performance of SHW systems were studied. The results showed that optimal design and control strategies based on the actual DHW profile can increase the annual solar energy utilization rate by 11.5%, and the corresponding electricity savings, CO₂ emission reduction, and cost saving are 163.10 kWh, 145.52 kg, and 81.06 RMB, per household. The location of the reheater had a significant influence on the energy performance and optimal measures of the SHW systems. Moreover, based on technologies or policies to adjust hot water demand, the water supply capacity of SHW systems can be matched with the hot water demand, which can improve the efficiency of solar energy utilization, save 8.5% of electrical energy for reheating, and lead to a 107.55 kg CO₂ reduction and 59.91 RMB cost saving per household per year.

1. Introduction

In the context of energy saving and emission reduction, China aims to peak CO₂ emissions before 2030 and carbon neutrality before 2060 [1]. Therefore, researchers have sought to reduce energy consumption and use clean energy. Domestic hot water (DHW) energy consumption accounts for 9.5% of the operational energy consumption of urban residential buildings in China [2]. Solar hot water (SHW) systems have been

widely used in residential buildings to save energy and reduce carbon emissions [3]. Twenty-one provinces in China require the installation of SHW systems in residential buildings, and the solar thermal utilization and collection area reached 800 million square meters by the end of 2020 [4].

Researchers have explored various methods to enhance the energy performance of equipment and SHW systems. Some scholars have focused on improving performance of solar collectors by using multi-layer selective solar absorbers [5]. Others have concentrated on

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Nomenclature			
A	reserve allowance percentage of the tank	P_t	probability of weekly shower times
ASHP	air source heat pump	Q_{CCCR_r}	traditional energy consumption per unit hot water of CCCR system
A_c	area of absorbing solar energy	Q_{CCTR_r}	traditional energy consumption per unit hot water of CCTR system
A_N	uniformly distributed pseudo-random number	$\dot{Q}_{load}$	rate at which energy is delivered to the taps in the form of hot water
CCCR system	centralized collective and centralized reheating system	R	solar radiation density
CCTR system	centralized collective and terminal reheating system	RMSE	root mean squared error
c_p	heat capacity of water	R_{use}	ratio of daily hot water consumption in evening
DHW	domestic hot water	SD	duration of shower
E_{CIR_P}	electricity consumption of circulating pump	SD_r	the remaining duration of the shower
E_{COL_P}	electricity consumption of collecting pump	SHW	solar hot water
E_{CR}	reheating energy of centralized-installed appliance	SST	shower start time
E_{eff_heat}	effective heat energy	ST_d	daily shower times
E_{pump}	electricity consumption of the pump	ST_w	weekly shower times
E_{p_loss}	heat loss of pipeline	t_{amb}	ambient air dry-bulb temperature
E_{reh}	reheating energy	t_{cc}	water temperature in collector at current time step
E_{SUP_P}	electricity consumption of supply pump	t_{cp}	water temperature in collector at previous time step
E_{t_loss}	heat loss of tank	$t_{c,in}$	temperature of water entering collector
E_{TR}	reheating energy of terminal-installed appliance	t_{pipe}	water temperature in pipeline
eff_{reh}	heating efficiency of reheating appliance	$t_{pipe,in}$	temperature of water that comes into pipe
F_{bot}	area of bottom layer exchanging energy with ambient air	$t_{pipe,out}$	temperature of water goes out pipe
F_c	area of solar panel exchanging heat with ambient air	t_R	water temperature of return water
F_{ex}	area of water layers having heat exchange	t_{req}	required temperature of hot water
F_{mid}	area of middle layer exchanging energy with ambient air	t_s	water temperature of supply water
F_{pipe}	area of pipeline exchanging energy with ambient air	$t_{t,bot}$	water temperature of bottom layer
F_{tank}	area of water tank	$t_{tc,bot}$	water temperature of bottom layer at current step
F_{top}	area of top layer exchanging energy with ambient air	$t_{tp,bot}$	water temperature of bottom layer at previous step
f_{CCCR}	solar fraction of CCCR system	$t_{t,mid}$	water temperature of middle layer
f_{CCCR_A}	annual solar fraction of the CCCR system	$t_{tc,mid}$	water temperature of middle layer at current step
f_{CCTR}	solar fraction of CCTR system	$t_{tp,mid}$	water temperature of middle layer at previous step
f_{CCTR_A}	annual solar fraction of the CCTR system	$t_{t,top}$	water temperature in top layer
G_C	flow of collector pump	$t_{tc,top}$	water temperature in top layer at current step
G_P	flow rate in pipeline	$t_{tp,top}$	water temperature in top layer at previous step
G_R	flow rate of return water in tank-to-consumer cycle	t_{use}	water temperature used by occupants
G_S	flow rate of water supply pump	VO	mass exchange volume between layers per time-step
G_T	flow rate of cycling pump	VO_{in}	volume into middle layer from near layers
I_o	intensity of “other”	VO_{out}	volume to near layers from middle layer
I_s	intensity of “shower”	V_c	volume of solar collector
K_{amb}	heat transfer coefficient of water to ambient air	V_{op}	capacity of water tank optimized based on actual hot water usage profile
K_c	heat transfer coefficient of the solar panel	V_{reh}	water volume needed to be reheated
K_{ex}	heat transfer coefficient between water layers	$V_{t,bot}$	water volume in bottom layer
K_{pipe}	heat transfer coefficient of pipeline from water to ambient air	$V_{t,mid}$	water volume in middle layer
L	total energy of hot water delivered to users	$V_{tc,top}$	water volume in top layer at current step
M	maximum available percentage of water volume without mixture	$V_{tp,top}$	water volume in top layer at previous step
N	number of users in this system	V_{use}	DHW usage amount
n	Regulated building lifetime limit	W	water use quota per capita per day
PRP	Poisson rectangular pulse	η_c	solar absorption efficiency of collector
P_d	probability of duration of shower	ρ	density of water
P_i	probability of intensity of hot water usage	τ_{pump}	operation duration of the pump
P_o	probability of “others”	τ	current time
P_{pump}	rated power of the pump	$\Delta\tau$	number of seconds per time step
P_s	probability of “shower”		

enhancing solar storage by applying phase change materials and techniques in thermal storage tanks, such as using stearic acid/palmitic acid as phase change materials [6], and packed bed thermal energy storage devices [7]. In addition to improving the performance of the equipment, many researchers have studied the optimal design form and operation of

systems, such as optimizing the SHW system operation schedules based on reinforcement learning [8], and proposed a new absorption solar thermal storage system with enhanced energy storage density from a double-stage output [9]. Moreover, SHW systems can be operated in tandem with other systems to achieve better energy performance, such

as integrating the systems with photovoltaic and solar thermal collector modules to make full use of solar energy [10], combining the systems with an air source pump (ASHP) to solve the discontinuous operation of the SHW system and frosting problem of ASHP [11], reducing the energy consumption of ASHP [12], and operating the SHW system with a ground source heat exchanger to form a desiccant system, which can provide thermal comfort and consume less energy [13].

Meanwhile, several researchers have reported that DHW usage behavior has a significant influence on the energy performance of SHW systems [14]. Various factors influencing DHW usage behavior in residential buildings were investigated via several surveys and measurements to further enhance our understanding. Hot water usage behavior varies according to region, climate, season, work or rest day, and time of day [15]. Occupants would use more hot water in regions with colder climates, particularly in winter and off-days. In a given day, hot water usage is generally concentrated in the morning or at night [15]. Further research revealed that socioeconomic demographic factors, such as occupancy, gender, age, number of generations, household income, degree of education, household size, consciousness of consumption, energy conservation, and environmental protection, also influenced DHW usage behavior in households [16].

For SHW systems, there are some special influencing factors, such as weather and SHW system performance (available water volume and hot water temperature). Sunny days, which usually correspond to a large amount and high temperature of available hot water in the SHW system, make occupants feel the hot water is “free” and “endless,” leading to excess hot water usage [17].

Several DHW profiles and models have been proposed to describe the usage behavior in residential buildings; the influence of DHW usage behavior on SHW systems has also been studied. Spur et al. [14] developed three realistic daily profiles of DHW usage based on measured data from Switzerland and Germany and simulated the SHW system. The results demonstrated that the DHW usage behavior significantly influenced the stored energy of hot water in the tank, and the largest difference under different DHW profiles was 13.3%. Edwards et al. [18] conducted a survey of 73 households in Quebec City, Canada, and prepared 12 annual DHW profiles, comprising four consumption levels and three temporal consumption patterns. In addition, they observed that the reheating energy, DHW load, and solar collection of SHW systems relied significantly on the DHW profiles. Jordan and Vajen [19] presented a stochastic Poisson rectangular pulse (PRP) DHW usage model that considered changes in DHW usage behavior during days, weekdays/weekends, and seasons. Subsequently, it was co-simulated with the SHW system model in TRNSYS. The results showed that the DHW load profile affected the fractional energy savings of the SHW system and

temperature stratification in the storage tank. Ahmed et al. [20] surveyed the DHW usage data of 379 occupants in 182 Finnish apartments from January 2012 to June 2014. Subsequently, monthly DHW profiles were proposed and co-simulated using an SHW system. The derived monthly profile showed a 4.7% higher delivered energy than that of the simulation without monthly variations.

The main DHW usage models are listed in Table 1. Currently, DHW behavior models mainly comprise rule-based and data-based models. Generally, mainstream rule-based models are stochastic models, whereas most data-based models are neural networks and time-series models. These DHW behavior models were generally established based on data surveyed from countries in North America and Europe, with a time resolution of 15 min.

Existing research has revealed that occupant behavior has a significant influence on the energy performance of SHW systems. However, this is a two-way effect. The operation information of SHW systems, such as water temperature and the available volume of hot water in the tank [17], also affects the hot water usage of the occupants. Although few tools can reflect the interaction between an occupant's DHW behavior and the performance of SHW systems, there is a lack of quantitative analysis for coordinating the occupants' water usage behavior and the design and operation of SHW systems to maximize the solar energy resource utilization, which can reduce energy consumption and CO₂ emissions.

In this study, based on survey data, an information-driven stochastic DHW usage model was proposed to reflect the occupants' stochastic DHW usage and the interaction with SHW systems. An information-driven module was embedded into the DHW behavior model to represent the influence of SHW system information on user behavior, which was used to reflect the actual phenomenon that occupants' behavior would be adjusted according to the information they received. The DHW behavior model and SHW sub-models were validated using measured data. Two types of SHW systems were studied: centralized collective and centralized reheating (CCCR) systems and centralized collective and terminal reheating (CCTR) systems. The optimal design and control strategy of SHW systems to match a user's actual DHW behavior was explored. In addition, the effects of occupants' sensitivity to water temperature and energy-saving awareness of occupants on the energy performance of SHW systems were studied. Moreover, the corresponding energy savings and CO₂ emission reductions for each optimal measure were calculated. By considering the user's behavior, the measures proposed in this study to improve solar energy utilization can effectively help in energy conservation and emission reduction in the building sector.

The remainder of this paper is organized as follows. Section 2

Table 1
Main existing DHW behavior models.

Types		Algorithms	Resolution	Data source
Rule-based	Stochastic	Activity-based model	1 min	UK [21]
			5 min	Sweden [22], UK [23]
			10 min	Canada [24]
			15 min	Germany [25]
			1 s	USA [26], Netherlands [27]
		PRP model	10 s	USA and Canada [28]
			1 min	Switzerland and Germany [19]
			6 min	USA [29]
			5 min	Sweden [30]
			1 h	Switzerland and Germany [31]
Data-based	Deterministic	Monte Carlo simulation models	5 min	Australia [32]
		Polynomial model	1 h	Finnish [33]
		Artificial neural network model	5 min	UK [34]
		Neural network model	1 h	Canada [35]
		Radial basic function neural network model	/	South Africa [36]
	Time series	Auto regressive moving average model	5 min	Romania [37]
			30 min	France [38]
			1 h	UK [39]
	SVM	Autoregressive integrated moving average model	1 h	China [40]
		Support vector machine model	1 h	

presents the technical approach, including the survey of the DHW behavior data, measurement of the SHW system, model establishment, model interaction and validation, and description of study cases. The results and analysis are presented in Section 3. Sections 4 and 5 present the discussion and conclusions, respectively.

2. Methodology

2.1. Technical approach

The technical approach of this study is shown in Fig. 1.

Based on a detailed survey on DHW consumption of residents, information, such as hourly DHW usage profile, weekly shower times, duration of shower, and time to start showering was extracted to establish a stochastic DHW usage model. Meanwhile, previous researchers have found that SHW information influences the DHW usage of occupants, including the water temperature and the available volume of hot water in the tank [17]. Therefore, in the water usage model, an information-driven module was added to reflect the influence of SHW systems' information on occupants' hot water usage. In this module, the information provided by the SHW system, like the water temperature, number of people using water, etc., was taken as input. The output of the module was adjustment of user's DHW behavior influenced by SHW information. The information-driven stochastic DHW usage model was integrated with various SHW equipment modules to construct SHW system models, reflecting the interaction between DHW usage behavior and the SHW system performance. These modules were validated through the comparison between measured and simulated DHW consumption data and SHW energy consumptions. Furthermore, two types of SHW systems were studied, namely the CCCR and CCTR systems. Case studies were conducted to investigate the optimal design methods of the two SHW systems considering the user's actual DHW use behavior. The influence of occupants' sensitivity to water temperature and their awareness regarding energy saving on the performance of the two SHW systems were also studied.

2.2. Data collection

2.2.1. DHW usage behavior surveys

According to the climatic conditions of different regions and their influence on buildings, there are five climatic zones in China: severe cold, cold, hot summer and cold winter, temperate, and hot summer and warm winter zone. To ensure that the collected data represented the hot water usage of Chinese residents, the researchers conducted the survey in some cities in each climate region through a simple sampling method [41], and online and field questionnaires [42]. The questions included DHW usage behavior, hot water supply temperature, and types of DHW systems. A detailed questionnaire is shown in Appendix A. To ensure the quality of the data, the questionnaires were reviewed. Questionnaires with too short a filling time (online) and obviously perfunctory answers (online and field) were regarded as invalid. There were 962 valid questionnaires, and the ratios of male to female respondents were 49% and 51%, respectively. The surveyed data on DHW usage behavior were adopted in this study.

Considering the influence of different regions and seasonal changes, the occupants' DHW consumption varied between 31 and 47 L/ (capita $\times$ day). The DHW usage patterns included showering, washing face, kitchen work, washing clothes, cleaning the house, and others. The distribution of weekly shower times, shower start time, and shower duration are shown in Tables 2–4.

The measured daily DHW usage profile exhibited an “M” shape, as depicted in Fig. 2. It can be seen that the highest water usage in China peaked in 20:00–23:00, which differs from that in Northern America and Europe, where it peaked in 06:00–09:00 [33]. The reason for this phenomenon is that Chinese residents shower in the evening, whereas people in Northern America and Europe shower in the morning [43].

Table 2
Shower times of users in a week.

Times	1–2	3–4	7	>7
Probability	0.205	0.504	0.241	0.050

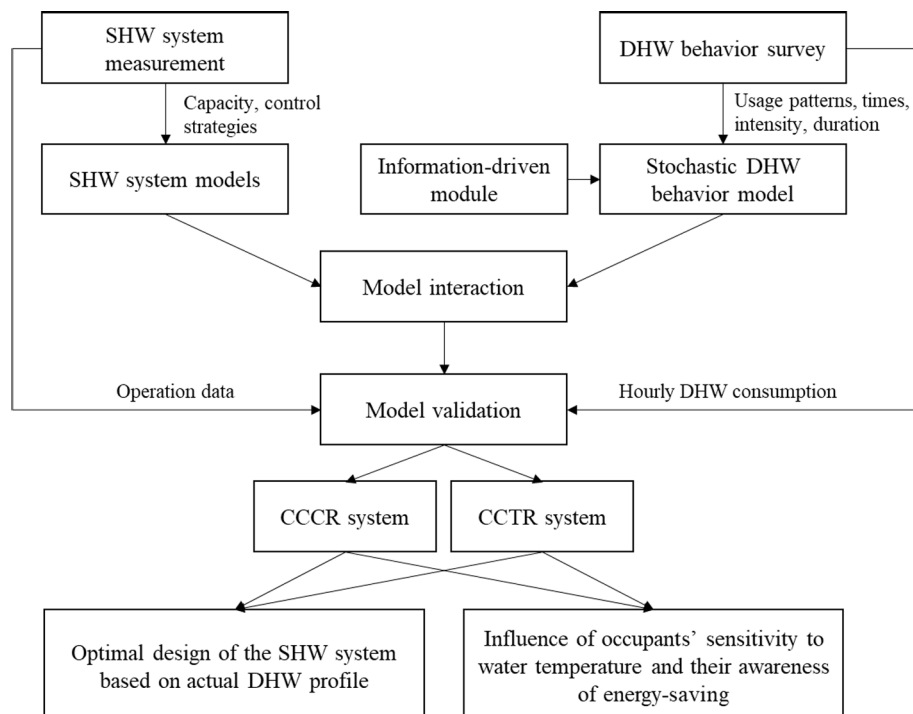


Fig. 1. Technical approach.

Table 3

Probability of shower start time.

Time	0:00–5:00	5:00–9:00	9:00–12:00	12:00–16:00	16:00–19:00	19:00–22:00	22:00–24:00
Probability	0	0.107	0.026	0.017	0.064	0.590	0.197

Table 4

Duration of shower (min).

Duration	5–10	10–20	>20
Probability	0.258	0.453	0.289

2.2.2. SHW system measurement

The field test for the SHW system was conducted from August 27 to September 1, 2014, in Chifeng, China. The measured SHW system was a CCCR hot water supply system for 90 users, as illustrated in Fig. 3.

This system comprises a solar collector, water storage tank, collecting pump, circulating pump, supply pump, water reheater, and pipelines. The collecting pump was on when the temperature difference between the solar collector and water tank was greater than 7 °C, and it turned off when the temperature difference was less than 3 °C. The circulating pump was kept on. The supply pump was used to fill the water tank up to 2/3rd of the total capacity when the water level was below 1/3rd of the total capacity. In addition, the supply pump filled the water tank at midnight every day. Meanwhile, to ensure that users could access hot water at any time, the water reheater maintained a minimum water temperature of 45 °C in the tank.

2.3. Information-driven stochastic DHW usage model

2.3.1. Probability functions for DHW usages

Owing to its effectiveness in simulating DHW profiles, the PRP model proposed by Buchberger and Wells [26] was adopted to describe the DHW behavior [27]. The input parameters of the PRP model included the number of users, water usage patterns, daily water usage times, and water usage duration and intensity. A profile with stochastic probability could better evaluate the energy efficiency of the SHW system than the repeated daily DHW usage profile [18]. Therefore, the stochastic characteristics of occupants' DHW usage were expressed by five probability functions: the probability of weekly shower times (P_t), shower start time (P_s), duration of shower (P_d), "other" hot water usage (P_o), and intensity of hot water usage (P_i).

Based on the data presented in Table 2, P_t follows the Poisson distribution expressed in Eq. (1).

$$P_t = \frac{3.98^{ST_w}}{ST_w!} e^{-3.98}, \quad ST_w = 0, 1, \dots \quad (1)$$

Based on the surveyed data, 91% of occupants use DHW for showers, accounting for approximately 80% of the DHW usage [42]. In this study, the DHW behavior was divided into two patterns: "shower" and "others"

For P_s , according to the data in Table 3, one day was divided into three time periods, and two normal distribution models were selected to describe p at different times (τ), as shown in Eq. (3). Following a uniform distribution, the generated weekly shower times (ST_w) were assigned to all days in a week to form the weekly shower schedule. Thus, the daily shower time (ST_d) was obtained. For each user, a uniformly distributed pseudo-random number (A_N) between 0 and 1 was generated and applied as an input to the inverse function of the cumulative probability function of P_s to find the corresponding shower start time (SST).

$$P_s = \begin{cases} p, & ST_d > 0 \\ 0, & ST_d = 0 \end{cases} \quad (2)$$

$$p = \begin{cases} 0, & 0:00 < \tau < 5:00 \\ 0.15 \left(\frac{1}{7\sqrt{2\pi}} e^{-\frac{(\tau-80)^2}{98}} \right), & 5:00 < \tau < 14:00 \\ 0.85 \left(\frac{1}{16\sqrt{2\pi}} e^{-\frac{(\tau-234)^2}{512}} \right), & 14:00 < \tau < 24:00 \end{cases} \quad (3)$$

The P_d , applied to generate the duration of shower (SD , min), followed a normal distribution, which was fitted based on the data in Table 4, as shown in Eq. (4). The duration of "others" pattern was 5 min by default.

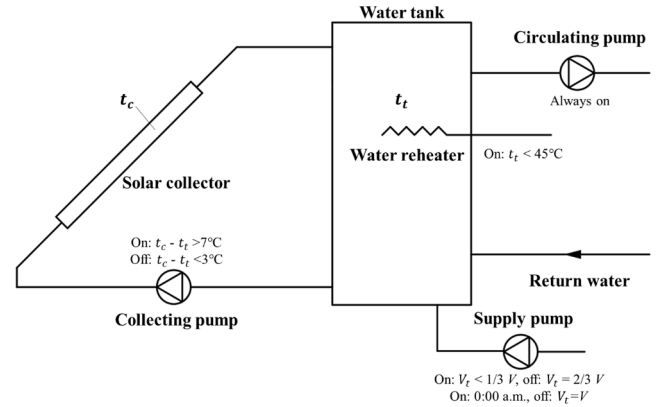


Fig. 3. Schematic operation strategy of CCCR system, Note: t_c : water temperature in the collector, t_t and V_t : water temperature and volume in tank, respectively; V : total capacity of tank.

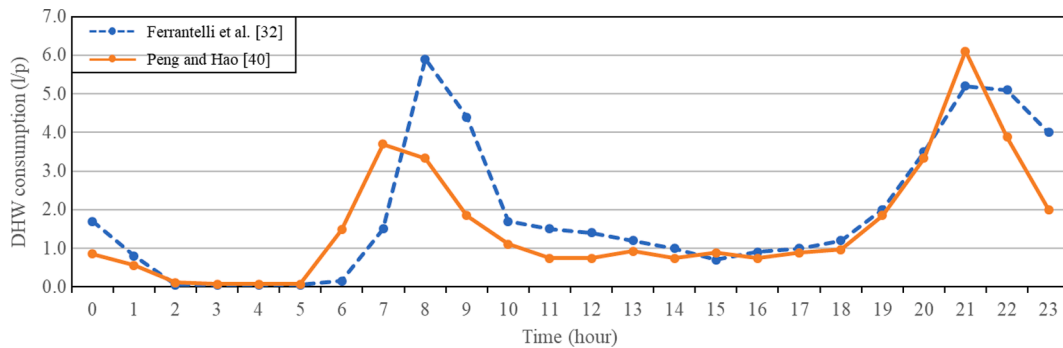


Fig. 2. Measured daily DHW usage profile and the comparison with the profiles in Northern America and Europe.

$$P_d = \frac{1}{8.18\sqrt{2\pi}} \exp\left(-\frac{(SD - 15.31)^2}{133.82}\right) N(15.31, 66.91) \quad (4)$$

Considering the DHW usage profile and probability of shower start time in Table 3, the P_o in different periods was obtained by fitting the profile in Fig. 2, as shown in Eq. (5).

$$P_o = \begin{cases} 0.05\%, & 0:00 < \tau < 5:00 \\ 1\%, & 5:00 < \tau < 24:00 \end{cases} \quad (5)$$

A comparison of the cumulative probabilities of P_t , P_s , and P_d with the surveyed data in Tables 2 to 4 is shown in Fig. 4, and the root mean squared error (RMSE) was applied to evaluate the accuracy of the fit functions. The results showed that the simulated probabilities were consistent with the measured data, and the RMSEs of P_t , P_s , and P_d were all less than 6.0%.

Based on the corresponding standard in China [44] and the survey data, the P_i also followed the normal probability distribution, including intensity of “shower” (I_s , L/min) and “others” (I_o , L/min), as shown in Eqs. (6) and (7), respectively:

$$I_s = \frac{1}{\sqrt{\pi}} \exp\left(-\frac{(u - 3.6)^2}{1}\right) N(3.6, 0.5) \quad (6)$$

$$I_o = \frac{\sqrt{5}}{\sqrt{2\pi}} \exp\left(-\frac{(u - 0.5)^2}{0.4}\right) N(0.5, 0.2) \quad (7)$$

2.3.2. Stochastic DHW usage model

Based on the five stochastic processes, an information-driven stochastic model of occupant DHW usage with a time resolution of 5 min was established, as shown in Fig. 5. The SHW system's information, including the water temperature and the available volume of hot water in the tank, would influence the DHW usage of occupants [17]. Therefore, an information-driven module was proposed and embedded into the DHW behavior model to represent the influence of the SHW system information on user behavior; it also reflects the adjustment of the user's DHW behavior influenced by SHW information, as indicated by the blue dashed box in Fig. 5. This module was used to reflect the actual phenomenon that residents would adjust their water usage time or amount according to the information provided by the SHW system. Meanwhile, the probabilities of different users responding to external information are different [45]. Therefore, residents' enthusiasm for responding to SHW information was also considered in the model and characterized by “response probability”

In the initial setting, the simulation period, the number of users, the ratio of responding users to all users and their response probability should be determined. At each time step, for each occupant, based on the τ and SD , the DHW usage amount was determined as follows:

- (1) At the beginning of each week, the ST_w was generated according to the P_t , and then ST_w was randomly assigned to seven days to form a weekly shower schedule. At the beginning of each day, the ST_d was renewed according to the schedule, and the SST was determined by P_s .
- (2) The current time (τ) was compared with SST . If $\tau = SST$, the user would take a shower. The SD was the duration of the shower, which was obtained from Table 4. In the calculation process, first, the total SD for a shower activity was generated through the function P_d . Then, the SD is allocated in the next few time steps. An intermediate variable SD_r was used to mark the remaining duration of the shower, which was equal to the SD at the beginning. For each time step, “ $SD_r > 0$ ” indicated that the user was still taking a shower, SD_r would be updated (by subtracting one time step ($\Delta\tau$) from itself) and passed to the next time step, and P_i was used to generate “shower” usage amount. “ $SD_r = 0$ ” meant that during this time step, the user would not take a shower. If $\tau \neq SST$, P_o was applied to generate “other” hot water usage pattern. Meanwhile, to prevent “shower” activities from happening too frequently, the minimum interval between the two showers was set to 6 h.
- (3) For users who respond to the “SHW information,” an information-driven module was enabled. If the “SHW information” met the user's requirement, a A_N was generated and compared with the users' response probability to determine the hot water usage.

2.4. Submodel of SHW systems

SHW systems mainly comprise solar collectors, water storage tanks, pipelines, pumps, and water reheaters. Through flexible combination and adjustment, these submodel can form various types of SHW systems.

2.4.1. Solar collector

The equations for calculating the water temperature in the collector vary according to the working conditions of the collecting pump. When the collecting pump is off, the collector absorbs solar energy and exchanges energy with the ambient environment without exchanging any energy or mass with the water storage tank. The equation is as follows:

$$c_p \rho V_c (t_{cc} - t_{cp}) = \eta_c R A_c \Delta\tau + K_c F_c \left(t_{amb} - \frac{t_{cc} + t_{cp}}{2} \right) \Delta\tau \quad (8)$$

where c_p and ρ denote the heat capacity and density of the water, respectively. t_{cc} and t_{cp} represent the water temperature in the collector at the current and previous time steps, respectively. V_c and η_c are the volume and solar absorption efficiency of the collector, respectively. R , A_c , $\Delta\tau$, K_c , F_c , and t_{amb} represent the solar radiation density, area of solar energy absorption, number of seconds per time step, heat transfer co-

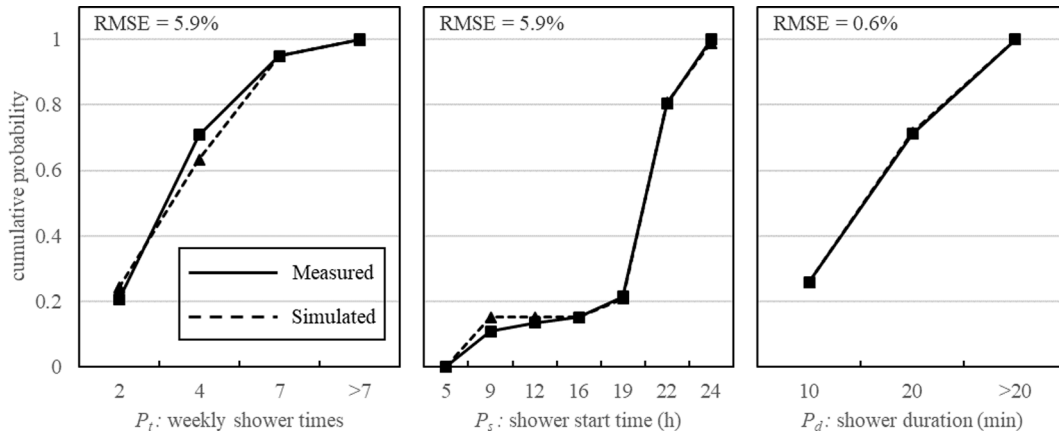


Fig. 4. Comparison of the DHW usage behavior functions (P_t , P_s , and P_d) with surveyed data.

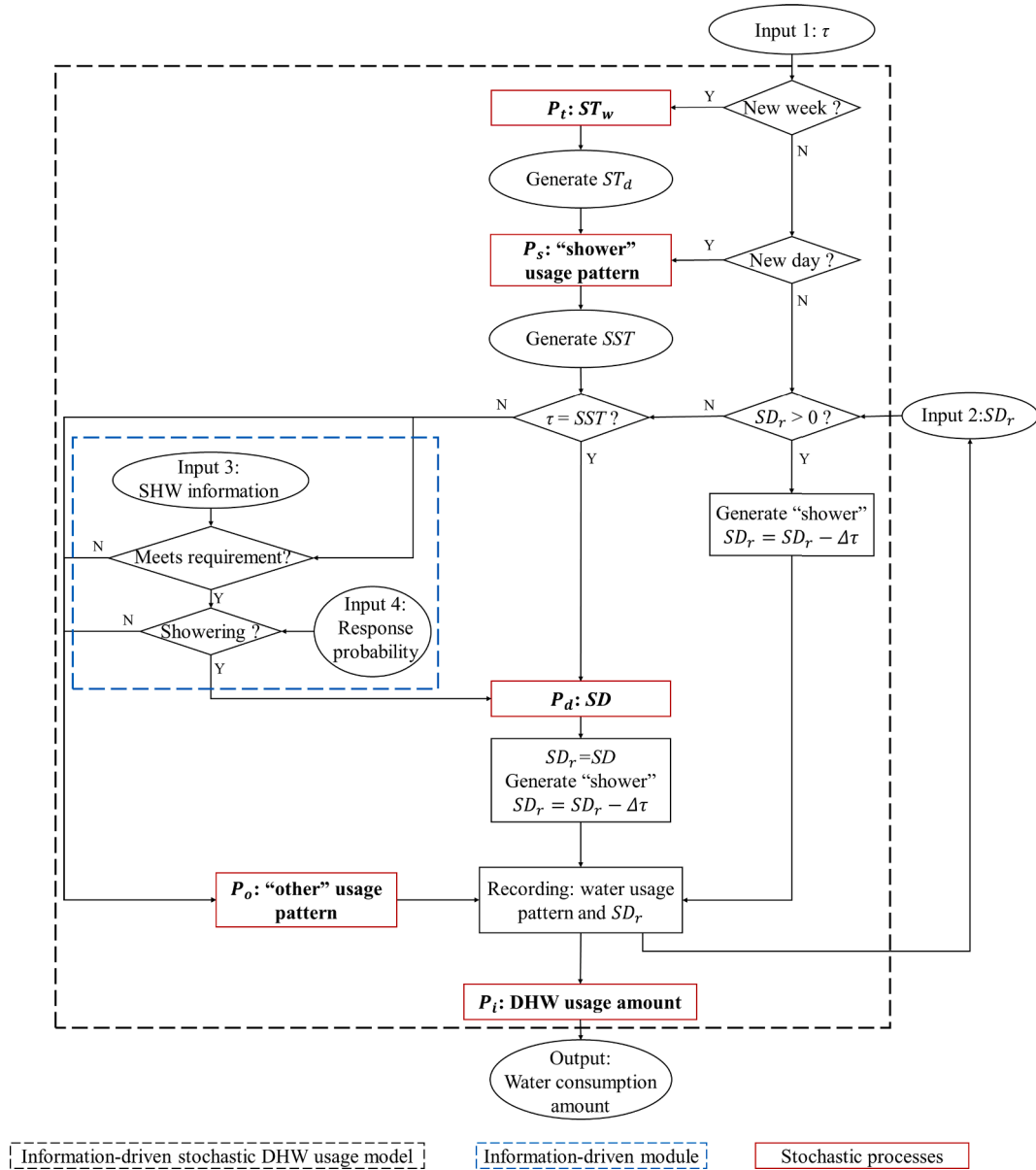


Fig. 5. Overview of information-driven DHW usage behavior model.

efficient of the solar panel, area of the solar panel exchanging heat with the ambient environment, and ambient air dry-bulb temperature, respectively.

When the collecting pump is on, the hot water in the collector flows into the tank, and mass and energy exchange occurs between the collector and tank. This process can be represented as follows:

$$c_p \rho ((V_c + G_C)t_{cc} - V_c t_{cp}) = G_C t_{c,in} c_p \rho + \eta_c R A_c \Delta \tau + K_c F_c \left(t_{amb} - \frac{t_{cc} + t_{cp}}{2} \right) \Delta \tau \quad (9)$$

where G_C and $t_{c,in}$ denote the flow of the collecting pump and the temperature of the water entering the collector, respectively.

2.4.2. Water storage tank

A multi-node water tank model was employed to describe the temperature stratification in a water storage tank [46]. Oberndorfer et al. [47] modeled water tanks with 1 to 100 nodes and proved that three to five nodes were sufficient. Therefore, the water tank model in this study was partitioned into five layers: bottom, three middle, and top layers.

The mass exchange volume of each layer at each time step was determined using the bottom layer, as shown in Eq. (10); it was delivered through the middle layers and reflected the fluctuation in the water level of the top layers.

$$V_0 = G_S + G_R - G_C \quad (10)$$

where V_0 denotes the mass exchange volume between layers per time step, G_S and G_R represent the flow rates of the supply pump and return water in the tank-to-consumer cycle, respectively.

The water temperature in the bottom layer was determined using Eqs. (11) and (12), respectively:

$$\begin{aligned} \rho c_p ((V_{t,bot} + G_S + G_R)t_{t,bot} - V_{t,bot}t_{tp,bot}) \\ = (G_R t_R + G_S t_S) \rho c_p + K_{amb} F_{bot} (t_{amb} - t_{t,bot}) \Delta \tau + K_{ex} F_{ex} (t_{t,mid} - t_{t,bot}) \Delta \tau \end{aligned} \quad (11)$$

$$t_{t,bot} = (t_{tp,bot} + t_{tc,bot})/2 \quad (12)$$

where $V_{t,bot}$ denotes the water volume in the bottom layer; $t_{t,bot}$ and $t_{tp,bot}$

represent the water temperature of the bottom layer at the current and previous steps, respectively. t_R , t_s , and $t_{t,bot}$ are the water temperatures of the return water, supply water, and bottom layer, respectively. K_{amb} , F_{bot} , K_{ex} , F_{ex} , and $t_{t,mid}$ are the heat transfer coefficient of water to ambient air, area of bottom layer exchanging energy with ambient air, heat transfer coefficient between water layers, area of water layers with heat exchange, and water temperature of the middle layer, respectively.

With the fluctuation in the water level, there are one to three middle layers in the tank, and the mass conservation and water temperature equations of the middle layers are shown in Eqs. (13)–(15).

$$V_{0,in} = V_{0,out} \quad (13)$$

$$\rho c_p ((V_{t,mid} + V_{0,in})t_{t,mid} - V_{t,mid}t_{t,bot}) = \rho c_p V_{0,in}t_{t,bot} + K_{amb}F_{mid}(t_{amb} - t_{t,mid})\Delta\tau + K_{ex}F_{ex}(t_{t,bot} + t_{t,top} - 2t_{t,mid})\Delta\tau \quad (14)$$

$$t_{t,mid} = (t_{t,bot} + t_{t,top})/2 \quad (15)$$

where $V_{0,in}$ and $V_{0,out}$ indicate the volume flow into the middle layer from adjacent layers and the volume flow to the adjacent layers from the middle layer, respectively. $t_{t,mid}$ and $t_{t,bot}$ represent the water temperature of the middle layer at the current and previous steps, respectively. $V_{t,mid}$, F_{mid} , and $t_{t,top}$ are the water volume of the middle layer, the area of the middle layer exchanging energy with the ambient environment, and water temperature of the top layer, respectively.

For the top layer, receiving water from the collector and middle layers and providing it to users, the equations determining the water temperature and volume are expressed as follows:

$$\begin{aligned} \rho c_p ((V_{t,top} + G_C + V_0)t_{t,top} - V_{t,top}t_{t,bot}) \\ = (G_C t_{cc} + V_0 t_{t,bot})\rho c_p + K_{amb}F_{top}(t_{amb} - t_{t,top})\Delta\tau + K_{ex}F_{ex}(t_{t,mid} - t_{t,top})\Delta\tau \end{aligned} \quad (16)$$

$$t_{t,top} = (t_{t,bot} + t_{t,mid})/2 \quad (17)$$

$$V_{t,top} = G_C + V_0 - G_T + V_{t,bot} \quad (18)$$

where $V_{t,top}$ and $V_{t,bot}$ represent the volume of water in the top layer at the current and previous steps, respectively; $t_{t,top}$ and $t_{t,bot}$ indicate the water temperature of the top layer at the current and previous steps, respectively. F_{top} , $t_{t,bot}$, and G_T are the areas of the top layer exchanging energy with the ambient environment, water temperature of the top layer, and flow rate of the circulating pump, respectively.

2.4.3. Pipeline

The equation used for the pipeline is expressed as follows:

$$G_p \rho c_p (t_{pipe,in} - t_{pipe,out}) = K_{pipe} F_{pipe} \Delta\tau (t_{amb} - \frac{t_{pipe,in} + t_{pipe,out}}{2}) \quad (19)$$

where G_p denotes the flow rate in the pipeline, $t_{pipe,in}$ and $t_{pipe,out}$ are the temperatures of water flowing into and out of the pipe, respectively. K_{pipe} and F_{pipe} denote the heat transfer coefficient of the pipeline from water to the ambient environment and the area of the pipeline exchanging energy with ambient environment, respectively.

2.4.4. Pumps

The pumps in the SHW system were considered as constant-frequency pumps, and the electricity consumption was calculated as follows:

$$E_{pump} = P_{pump} \times \tau_{pump} \quad (20)$$

where E_{pump} , P_{pump} , and τ_{pump} represent the electricity consumption, rated power, and operation duration of the pump, respectively.

2.4.5. Water reheater

In this study, the reheater capacity was assumed to be infinite, which indicates that the reheater could achieve the required temperature in one time step. The required water temperature was 45 °C [44]. The heating efficiency of the reheater was assumed to be 100%. The reheating energy of the reheater is expressed as follows:

$$E_{reh} = \begin{cases} \frac{(t_{req} - t_i)\rho c_p V_{reh}}{eff_{reh}}, & \text{if } t_i < t_{req} \\ 0, & \text{if } t_i \geq t_{req} \end{cases} \quad (21)$$

where E_{reh} , t_{req} , V_{reh} , and eff_{reh} denote the reheating energy, required hot water temperature, water volume needed to be reheated, and heating efficiency of the reheating appliance, respectively.

2.5. Model interaction

Two models (i.e., DHW usage and SHW system models) interacted with the hot water usage amount. At each step, the information-driven stochastic DHW usage model provided the total hot water usage amount to the SHW system model. Meanwhile, the SHW information would influence the occupant's DHW behavior, and users had a probability of using hot water when the signal from the SHW system met their requirements. The interaction between the information-driven stochastic DHW usage and SHW system models is shown in Fig. 6.

2.6. Model validation

The measured and simulated DHW usage profiles are compared in Fig. 7. This shows that the distribution of the simulated DHW usage profile is consistent with the survey results. The average simulated daily DHW consumption (38.6 L/capita) was close to the measured data (31–47 L/capita, average: 37 L/capita), which proved the effectiveness of the stochastic DHW usage model.

The measured SHW system performance data in Section 2.2.2, was used to verify the accuracy of the SHW sub-models. The equipment capacity, operation control, and simulation period were set according to the measured CCCR system. Three simulated results, namely the heat loss of the pipeline ($E_{p,loss}$), effective heat energy ($E_{eff,heat}$), and heat loss of the tank ($E_{t,loss}$), were compared with the measured data. These values were calculated as follows:

$$E_{p,loss} = (t_{pipe} - t_{amb})K_{pipe}F_{pipe}\Delta\tau \quad (22)$$

$$E_{eff,heat} = (t_{use} - t_s)\rho c_p V_{use}\Delta\tau \quad (23)$$

$$E_{t,loss} = (t_i - t_{amb})K_{amb}F_{tank}\Delta\tau \quad (24)$$

where t_{pipe} and t_{use} represent the water temperature present in the pipeline and used by the occupants, respectively, and F_{tank} indicates the area of the water tank.

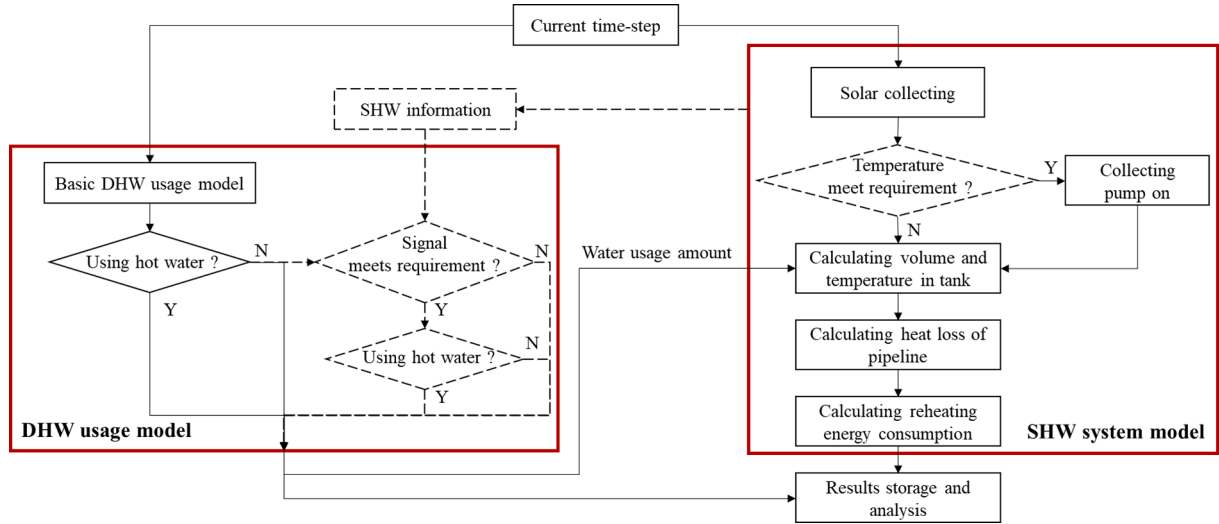


Fig. 6. Interaction between SHW system and DHW usage model.

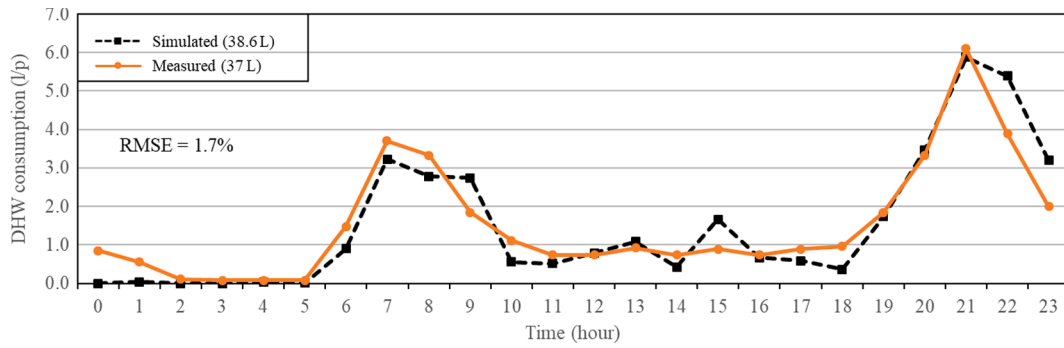


Fig. 7. Comparison between measured and simulated DHW usage profiles.

Table 5

Comparison of measured and simulation operation results of CCCR system.

Results	Measured (kWh)	Simulated (kWh)	Bias (%)
E_{p_loss}	288	261.0	-9.4
E_{t_loss}	45	43.5	-3.3
E_{eff_heat}	686	742.0	8.2

The simulated and measured data are listed in Table 5. The differences were within $\pm 10\%$, which demonstrates the effectiveness of these sub-models.

2.7. Case studies

The operation and performance of two types of SHW systems, namely, the CCCR and CCTR hot water supply systems, were investigated in this study. The CCTR system, which lowers the heat loss of the water tank, is an improvement based on the CCCR system [48]. A schematic illustration of the CCTR system operation is shown in Fig. 8.

There were two main differences between the CCCR and CCTR systems. The CCCR system has three pumps: collecting, circulating, and supply pumps, whereas the CCTR system uses gravity to supply water to the users instead of a circulating pump. The second major difference was that the water reheater in the CCCR system was installed in the water storage tank, whereas in the CCTR system, it was installed at the user terminals. The equations for the centralized and terminally installed reheaters are shown in Eqs. (25) and (26), respectively.

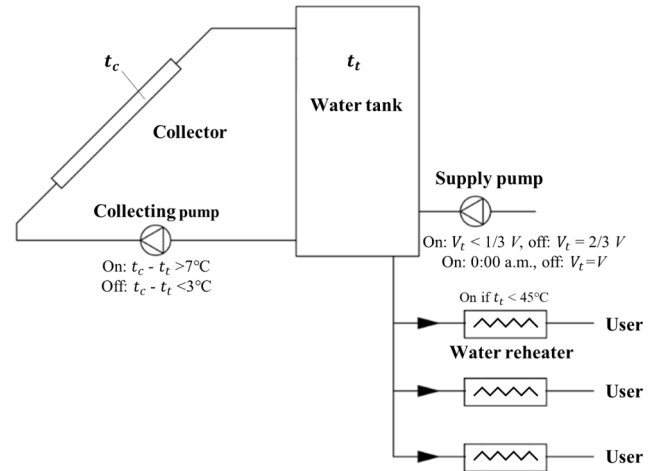


Fig. 8. Schematic illustration of CCTR system operation.

$$E_{CR} = \begin{cases} (t_{req} - t_t) \rho c_p V_t / eff_{reh}, & \text{if } t_t < t_{req} \\ 0, & \text{if } t_t \geq t_{req} \end{cases} \quad (25)$$

$$E_{TR} = \begin{cases} (t_{req} - t_t) \rho c_p V_{use} / eff_{reh}, & \text{if } t_t < t_{req} \\ 0, & \text{if } t_t \geq t_{req} \end{cases} \quad (26)$$

where E_{CR} and E_{TR} represent the reheating energies of the centralized and terminal-installed reheaters, respectively, and V_{use} indicates the

Table 6

Overview of main setting and research objective of the study cases.

Target	No	Settings and adjustment	Ratio of response users and response probability	Research Objective
Optimal design of SHW systems	1	According to the measured data Tank: $D = 2$ m, $H = 2.5$ m, $V = 7.85$ m ³ ; Pump _c : on: $\Delta T_{ct} > 7$ °C, off: $\Delta T_{ct} < 3$ °C; Pump _s : 0:00 a. m.: fill the tank; Other times: on: $V_t < 1/3$ V, off: $V_t = 2/3$ V	0 / 0	Basic case
	2	Reduce volume of water tank Tank: $D = 2$ m, $H = 1$ m, $V = 3.14$ m ³ ; Pump _c : on: $\Delta T_{ct} > 3$ °C, off: $\Delta T_{ct} < 1$ °C; Pump _s : 0:00 a. m.: fill the tank; 14:00–19:00: keep tank full; other times: on: $V_t < 1/5$ V, off: $V_t = 4/5$ V;	0 / 0	Based on the actual user water distribution to optimize the system to check whether the system can meet the user's heat demand
Interaction of occupants' behavior and the performance of SHW systems	3	Case 1 + water temperature feedback	90% / 90%	Explore impact of the information feedback on the standard designed SHW system
	4	Case 2 + water temperature feedback	90% / 90%	Investigate the influence of the information feedback on the adjusted SHW system

Note: D: diameter; H: height; Pump_c: collecting pump; Pump_s: supply pump; ΔT_{ct} : difference in water temperature between the solar collector and water tank.

DHW usage amount.

Based on the two SHW systems, four cases were studied, as listed in Table 6.

Case 1 was considered the basic case. The sizing of the system and operation strategy of the pumps were set according to the measured data. In this case, during the design process, only daily hot water consumption was considered.

In Case 2, the sizing of the tank capacity and the operation strategy of the pumps were improved according to the users' DHW water usage profile. This means that the heat storage should be increased before the peak night water use for the occupants using more water at night. Therefore, the supply pump would keep the amount of water in the tank full in the afternoon, and the collecting pump is set to turn on when the temperature difference between the collector and tank is over 3 °C, and off when it is below 1 °C. Thus, the SHW system can capture more solar energy before the peak energy consumption. In addition, to reduce the impact of the decline in solar energy utilization caused by water replenishment [49], the control measures of the supply pump were

adjusted as follows: at 14:00–19:00: tank kept full; other times: replenished water to 4/5 capacity when the water volume dropped to 1/5; and at midnight, the water tank was filled. Meanwhile, the water tank size was reduced to 3.14 m³, calculated using Eq. (27).

$$V_{op} = W \times N \times R_{use} \times (1 + A) / M \quad (27)$$

where V_{op} and W represent the capacity of the water tank optimized based on the actual hot water usage profile (m³) and water usage quota per capita per day (47 L, based on the survey). N , R_{use} , A , and M are the number of users in this system, the ratio of daily hot water consumption in the evening (50%), reserve allowance percentage of the tank (15%), and maximum available percentage of water volume without mixture (the value was 80% in this case according to the control of the supply pump), respectively.

Cases 3 and 4 were developed based on Cases 1 and 2, respectively. These two cases were designed to check the influence of occupants' sensitivity to water temperature and their awareness of energy saving on the performance of SHW systems. In these cases, the SHW systems provided information to the users regarding the water temperature in the tank. It was assumed that 90% of users might react when the water temperature in the tank reaches 55 °C, and the probability that these users use hot water (i.e., response probability) is 90%. The other settings of Cases 3 and 4 were the same as those of Cases 1 and 2, respectively.

For all four cases, based on the typical meteorological year (TMY) data of Chifeng [50], two days (January 11 and July 13) were selected as typical days for analysis, which represent days with high and low solar radiation intensities [51], as shown in Table 7.

3. Results and analysis

Based on previous research [52], the solar fraction was used to evaluate the energy-saving effect of SHW systems, as expressed in Eqs. (28)–(30). The higher the solar fraction, the greater is the solar energy utilized by the SHW system under the same hot water consumption, which indicates a better energy-saving and emission reduction effect.

$$f_{CCCR} = \frac{L - E_{CR}}{L} \quad (28)$$

$$f_{CCTR} = \frac{L - E_{TR}}{L} \quad (29)$$

$$L = \int \dot{Q}_{load} dt \quad (30)$$

where f_{CCCR} and f_{CCTR} represent the solar fractions of the CCCR and CCTR systems, respectively, L represents the total energy of hot water delivered to users, and $\dot{Q}_{load}$ denotes the rate at which energy is delivered to the taps as hot water.

3.1. Annual energy-saving effects

The annual solar fractions of the two systems for the four cases are listed in Table 8.

The annual solar fraction of the CCCR system ($f_{CCCR,A}$) in Case 2 was 11.5% higher than that of Case 1, which showed that considering the occupants' DHW usage profile to improve the system design and control strategies was a highly effective measure for increasing the utilization rate of solar energy.

Based on the survey data, assuming that the domestic hot water

Table 7

Two typical days chosen for analysis and corresponding solar radiation.

Solar radiation (MJ / (m ² × day))	Days selected/radiation (MJ / (m ² × day))
< 8	January 11th / 7.87
≥ 16	July 13th / 20.65

Table 8

Annual solar fraction of two systems in four study cases.

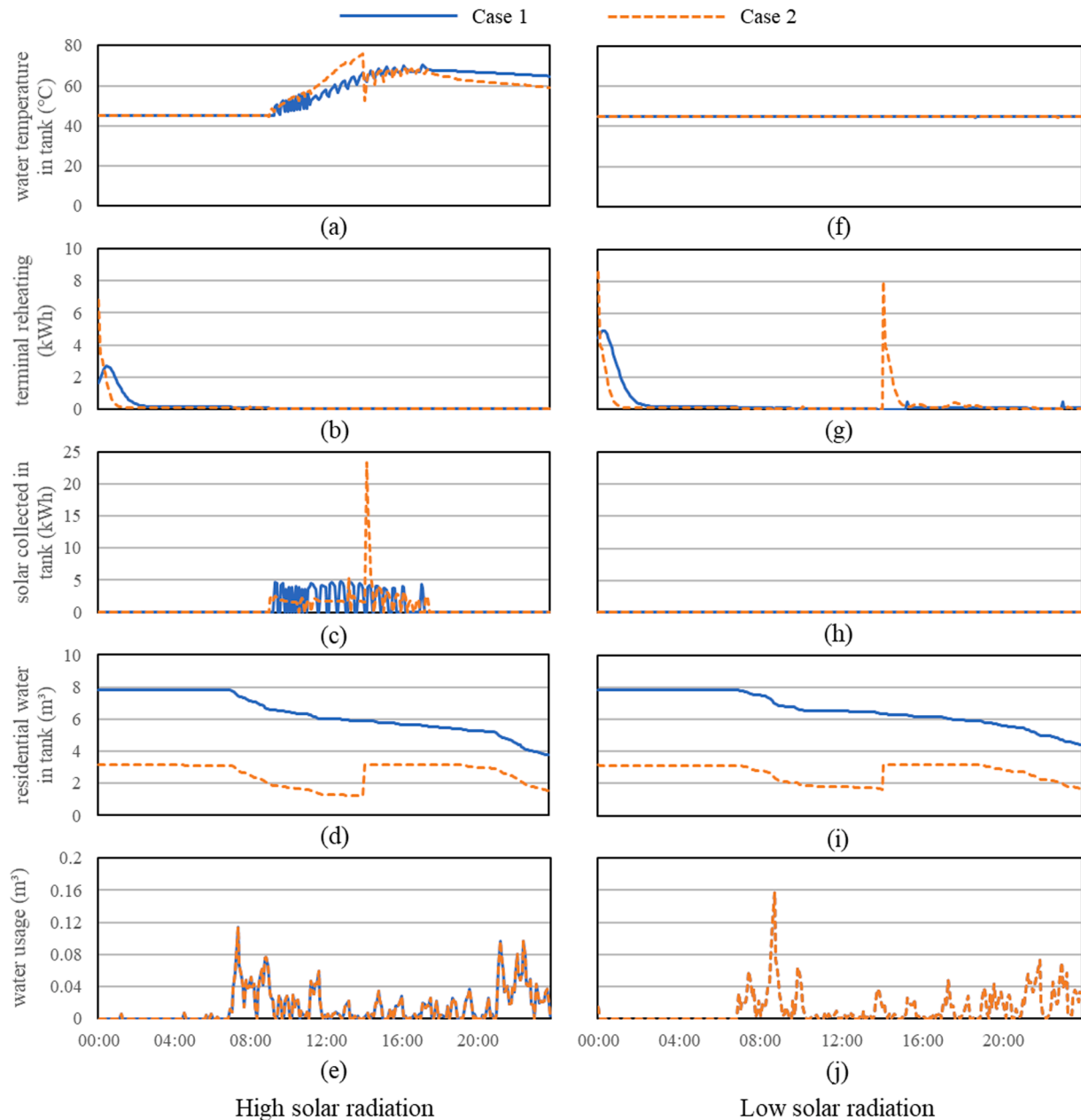
Systems	Annual solar fraction (%)			
	Case 1	Case 2	Case 3	Case 4
CCCR	24.3	35.8	24.3	41.6
CCTR	69.2	70.1	69.1	72.7

consumption was 111 L per household ($37 \text{ L}/(\text{capita} \times \text{day})$, 3 people in a household), and the temperature of the supplied cold water was 15°C and the required temperature of hot water was 45°C . Therefore, the usage amount of annual domestic hot water was 40.52 m^3 per household, and the demand energy of hot water was 1418.20 kWh per household. By increasing f_{CCCR_A} 11.5% higher, the reduction of annual

electricity on reheating water would be 163.10 kWh per household, and the corresponding cost saving was 81.06 RMB per household (the local electricity cost in Chifeng city is 0.497 RMB/kWh). Considering that the emission factor is $0.8922 \text{ kg CO}_2/\text{kWh}$ for the northwest grid [53], the annual reduction in CO_2 emissions in this case was 145.52 kg per household.

For the CCTR system, the annual solar fraction of the CCTR system (f_{CCTR_A}) of Cases 1 and 2 were similar, with slightly higher values for Case 2. Therefore, the differences between the two cases are insignificant from the perspective of increasing solar energy utilization. However, in Case 2, the capacity of the water tank is reduced. This can save the investment of the SHW system and have a positive effect on applications in buildings with limited available space for equipment.

From the comparison of f_{CCCR_A} and f_{CCTR_A} in Cases 1 and 3, it is



Note:

Case 1: standard designed CCCR system;

Case 2: optimized CCCR system based on actual DHW usage profile.

Fig. 9. Comparison of the operation of the standard designed and optimized CCCR system.

observed that for the standard designed SHW systems, water temperature feedback seems to have an insignificant effect on the SHW system performance. However, with the optimal design of the SHW system (presented in Cases 2 and 4), the water temperature signal feedback can play a significant role. Compared with Case 2, the improvements in f_{CCCR-A} and f_{CCTR-A} are 5.8% and 2.6%, respectively, in Case 4.

3.2. Operating profiles on typical days

3.2.1. Optimal design of SHW systems

Detailed operating profiles for the CCCR system are shown in Fig. 9.

During the days with high solar radiation, the water temperature from 0:00 to 08:00 in the tank for the two cases was the same. At 14:00, in Case 2, owing to the optimal control of the supply pump (14:00–19:00: tank kept full), the tank was filled up, as depicted in Fig. 9 (d), and the water temperature in the tank decreased to 52.46 °C. This triggered the collector to deliver hot water to the tank, resulting in a peak of solar collection (23.33 kWh), as shown in Fig. 9 (c), and the temperature in the tank increased to 61.99 °C. For the same period in Case 1, the water temperature in the tank increased from 45 to 62.19 °C, and that of the two cases rose to 67.11 °C at 15:00, as shown in Fig. 9 (a). During the rest of the day, in Case 1, constrained by the operation control of pumps, the system could not achieve a higher water temperature despite abundant solar energy availability. In Case 2, the collected solar energy was used to heat the replenished cold water; therefore, the water temperature in the water tank did not increase significantly. For the rest of the day, two cases had a sufficient amount of hot water above 58 °C; the SHW system in both cases provided sufficient hot water to occupants during the night hot water usage peak.

From these detailed operating profiles, it can be observed that the daily hot water usage for the standard designed CCCR system was less than 2/3 of the total capacity of the tank, as shown in Case 1. Therefore, starting from 00:00, water was not replenished throughout the day. The water refilled during the early morning was heated to 45 °C. Because no sun was present at this time, all the heating energy was provided by the reheater. It can also be observed from Fig. 9 (b) that electric heating mainly occurred during this period. However, for Case 2, owing to the reduced size of the water tank and optimal pump control strategy, the water replenishment was conducted twice throughout the day, one of which occurred at 2p.m. At this time, the solar energy was sufficient, and the heating energy required for water supplementation could be obtained from solar energy, as shown in Fig. 9 (c). Therefore, compared with Case 1, the electric supplementary heat in the early morning was reduced, and the total supplementary heat consumption of the system was also reduced. On July 13 (the typical day with high solar radiation), the solar fraction of Case 2 was 75.75% (Case 1: 64.94%). It can be concluded that on a typical day with high solar radiation, for the CCCR system, because the reheater was used to heat the amount of hot water in the tank, the smaller water tank volume led to less reheating energy consumption.

On days with low radiation (January 11, for instance), because there was reheating in the water tank, the water temperature difference between the solar collector and water tank was unable to reach the starting condition of the collecting pump for both cases. Therefore, the water tank failed to extract solar heat from the collector, and all the hot water energy came from the reheater, and the reheating energy consumption of the two cases was close (Case 1: 209.28 kWh, Case 2: 208.10 kWh).

Considering the differences in the circulating pumps of the two SHW systems, the traditional energy consumption per unit hot water (Q_r) [42] of the CCCR system (Q_{CCCR-r}) and CCTR system (Q_{CCTR-r}) were calculated to evaluate the coefficient of energy-saving performance, as shown in Eqs. (31) and (32). The smaller Q_r is, the more energy-saving the SHW system will be.

$$Q_{r-CCCR} = \frac{E_{CR} + E_{COL-P} + E_{CIR-P} + E_{SUP-P}}{V_{use}} \quad (31)$$

$$Q_{r-CCTR} = \frac{E_{CR} + E_{COL-P} + E_{SUP-P}}{V_{use}} \quad (32)$$

where E_{COL-P} , E_{CIR-P} , and E_{SUP-P} indicate the electricity consumption of the collecting, circulating, and supply pumps, respectively. The Q_{r-CCCR} in Case 1 were 17.66 kWh/m³ (high radiation) and 51.64 kWh/m³ (low radiation), whereas that in Case 2 were 12.71 kWh/m³ and 51.42 kWh/m³ respectively, which also further prove that the optimal strategy based on actual hot water usage profile would improve the energy-saving performance of the CCCR system.

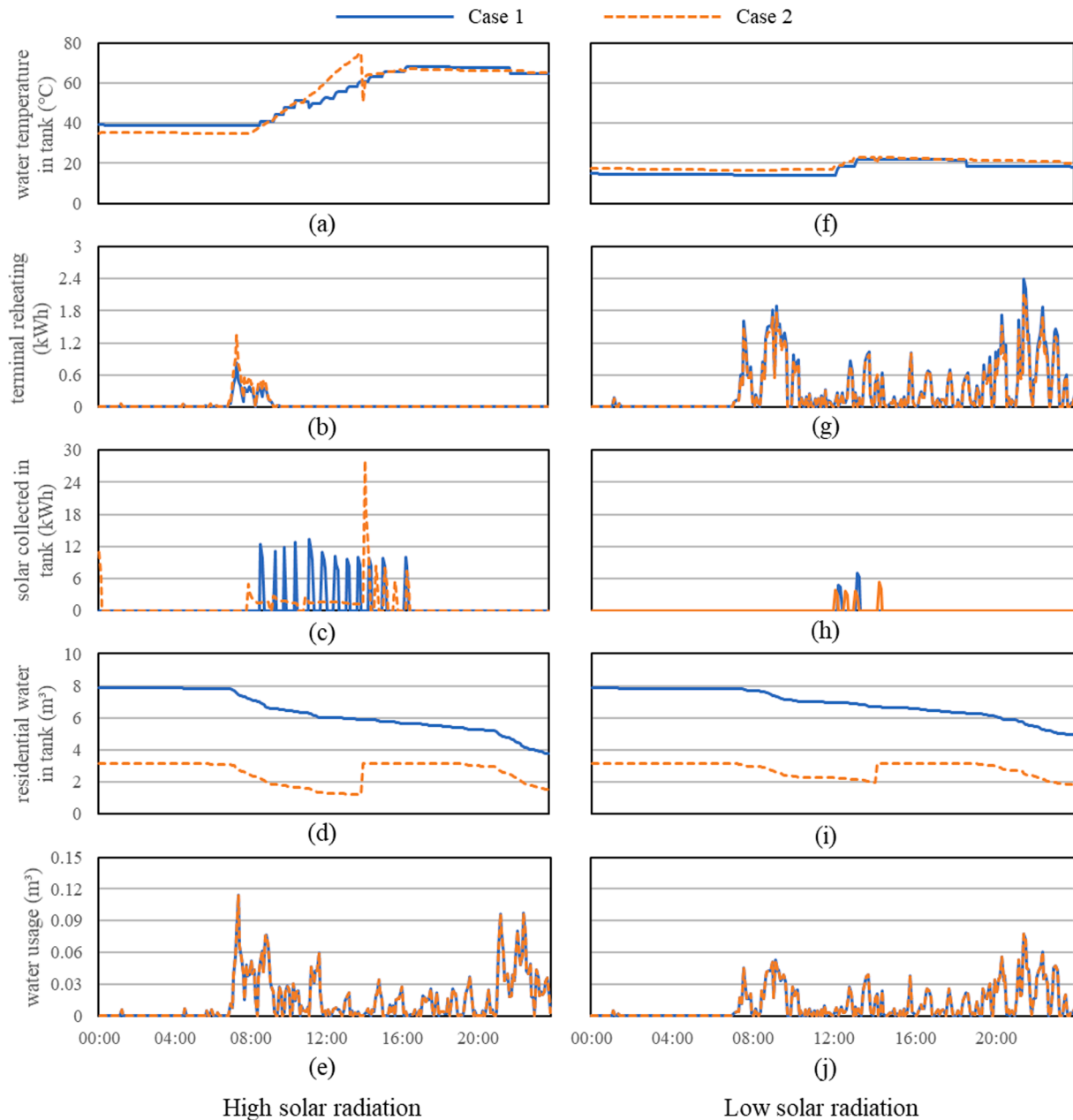
The detailed operating profiles of the CCTR system in Cases 1 and 2 are compared in Fig. 10. In contrast to the CCCR system, reheating occurred in the user terminals; therefore, the CCTR system did not need to maintain the water temperature in the tank above 45 °C. Hence, there was no reheating after water replenishment in the early morning, as shown in Fig. 10 (b). Meanwhile, it should be noted that because the water tank had a smaller capacity, the water temperature in the tank of Case 2 was significantly affected by water replenishment. As shown in Fig. 10 (a), the water temperatures in the tanks of the two cases were similar at the end of the day. The proportion of replenished water accounted for approximately 60% of the total tank capacity in Case 2, whereas it was approximately 50% in Case 1. Therefore, the water temperature in the tank in Case 2 was lower than that in Case 1 after water replenishment. Hence, from 0:00 to 9:30, as shown in Fig. 10 (b), a lower water temperature in the tank of Case 2 led to a higher reheating energy consumption (12.40 kWh) than that of Case 1 (8.33 kWh). From 10:00 to 24:00, the water temperature in both cases was higher than 45 °C, and there was no need for reheating. Therefore, for the CCTR system, it can be concluded that on a typical day with high solar radiation, because the reheater was used to heat the amount of hot water used by the occupant, the smaller water tank volume, on the contrary, led to more reheating energy consumption.

When the solar radiation was low, with no reheating in the water tank, the temperature difference between the solar collector and the water tank could meet the starting condition of the collecting pump. In particular, for Case 2, with a lower temperature difference requirement, the collecting pump could collect more solar energy (67.26 kWh) than that of Case 1 (45.05 kWh). In addition, owing to the small water tank capacity, the temperature rise in the tank in Case 2 was greater than that in Case 1 under the same solar radiation intensity, even after the daily water replenishment, as shown in Fig. 10 (f). Overall, on a typical day with low solar radiation, for the CCTR system, the solar fraction in Case 2 was 16.80%, which was 7.55% higher than that of Case 1.

For the CCTR system, in Case 1, the Q_{r-CCTR} was 2.26 kWh/m³ on the day with strong solar radiation and 31.54 kWh/m³ on the day with low solar radiation. The Q_{r-CCTR} in Case 2 (3.61 kWh/m³) was higher than that in Case 1 when the solar radiation was high, and that was smaller (28.24 kWh/m³) with low solar radiation. The results were consistent with the analysis of the application of the optimal strategy in the CCTR system.

From a detailed analysis of the operating profiles, it can be concluded that the installation location of the reheater had a significant influence on the selection of optimal measures for SHW systems. For the CCCR system, a water reheater was installed in the centralized water tank, which heated the water of the entire tank. The CCTR system was installed at the user terminals only to heat the amount of water used by the occupants. Therefore, in the CCCR system, the energy consumption of the reheater is not synchronized with the user's actual heat demand, whereas it is synchronized in the CCTR system. Hence, in the CCCR system, optimizing the design of the system according to the user's hot water usage behavior may compensate for the lack of synchronization between the reheat and demand, and can effectively improve the solar energy utilization. However, the same measure was ineffective for the CCTR system.

Meanwhile, it should be noted that in the CCTR system, the operation



Note:

Case 1: standard designed CCTR system;

Case 2: optimized CCTR system based on actual DHW usage profile.

Fig. 10. Comparison of the operation of the standard designed and optimized CCTR system.

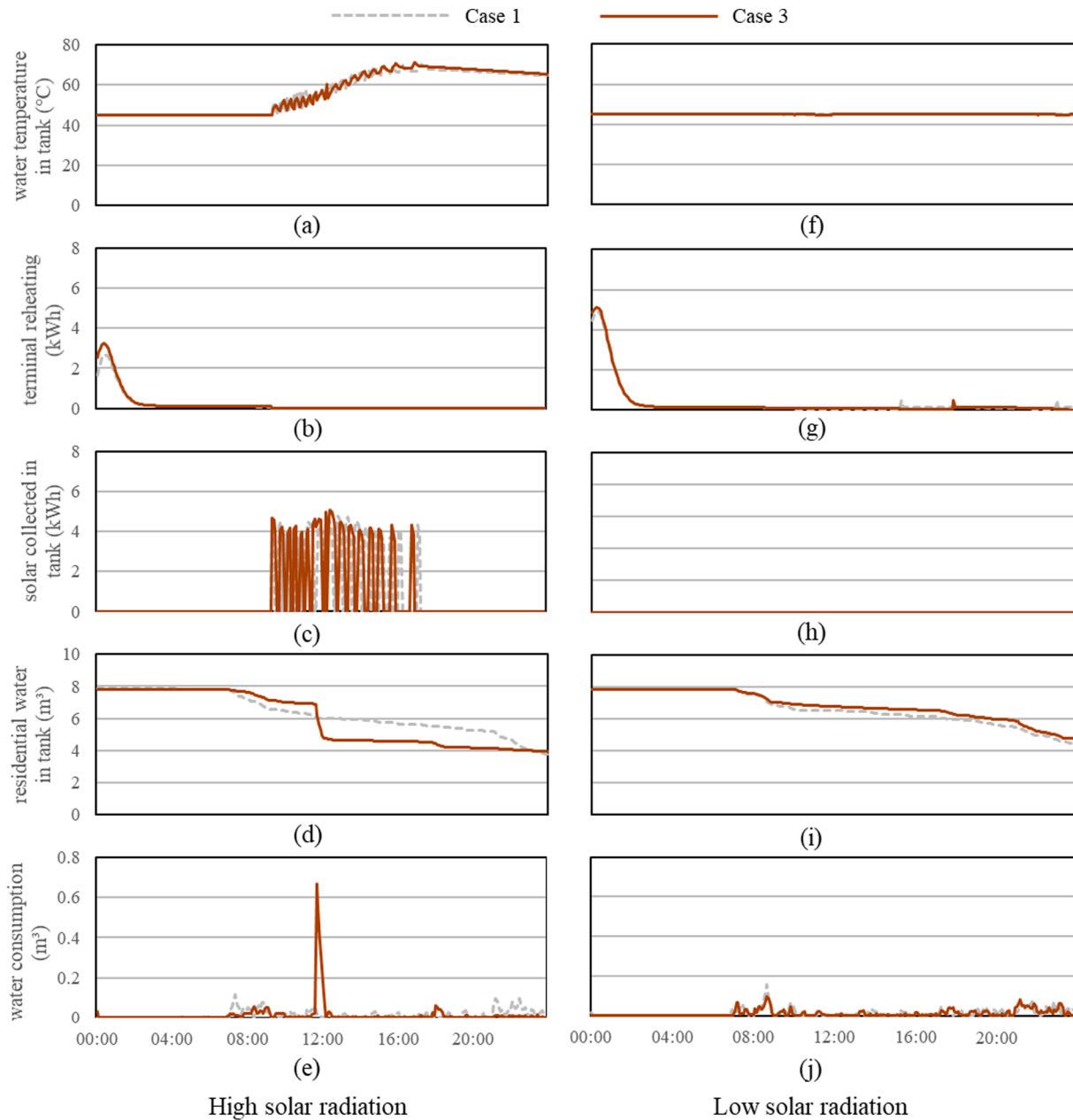
of the reheater matches the user's needs in time. This feature enables the water temperature to vary significantly. Therefore, a larger temperature difference between the water temperature in the tank and collector can be achieved through better use of solar energy. Consequently, it can be concluded that f_{CCTR_A} was generally greater than f_{CCCR_A} in all cases.

3.2.2. Interaction of occupants' behavior and the performance of SHW systems

Figs. 11 and 12 show that providing the water temperature in the tank to the occupants would not influence the solar energy utilization in the standard designed CCCR and CCTR systems.

Considering the CCCR system as an example, as shown in Fig. 11 (a), when the solar radiation was high, the water temperature in the tank reached 55 °C at 11:30. Subsequently, 90% of the occupants took a

shower immediately, which resulted in a peak in heat demand, as shown in Fig. 11 (e). However, owing to the large capacity of the water tank, the amount of water met the user's water demand, and there was no need to supplement the water, as shown in Fig. 11 (d). The supply pump did not start throughout the day, and Fig. 11 (a) shows that the water temperature was above 60 °C at the end of the day. During the days with low radiation, as shown in Fig. 11 (f), the water temperature in the tank did not reach 55 °C for the entire day; thus, the feedback module did not work. Although the information feedback adjusted the water usage peak from evening to noon in days with strong solar radiation, considering the entire day, the operation of the CCCR system did not change. Therefore, with a large water tank capacity, the hot water supply of the system was significantly greater than the user's demand; hence, the adjustment of occupants' water demand had a slight influence on the CCCR and CCTR



Note:

Case 1: standard designed CCCR system;

Case 3: standard designed CCCR system with feedbacking water temperature in tank to users.

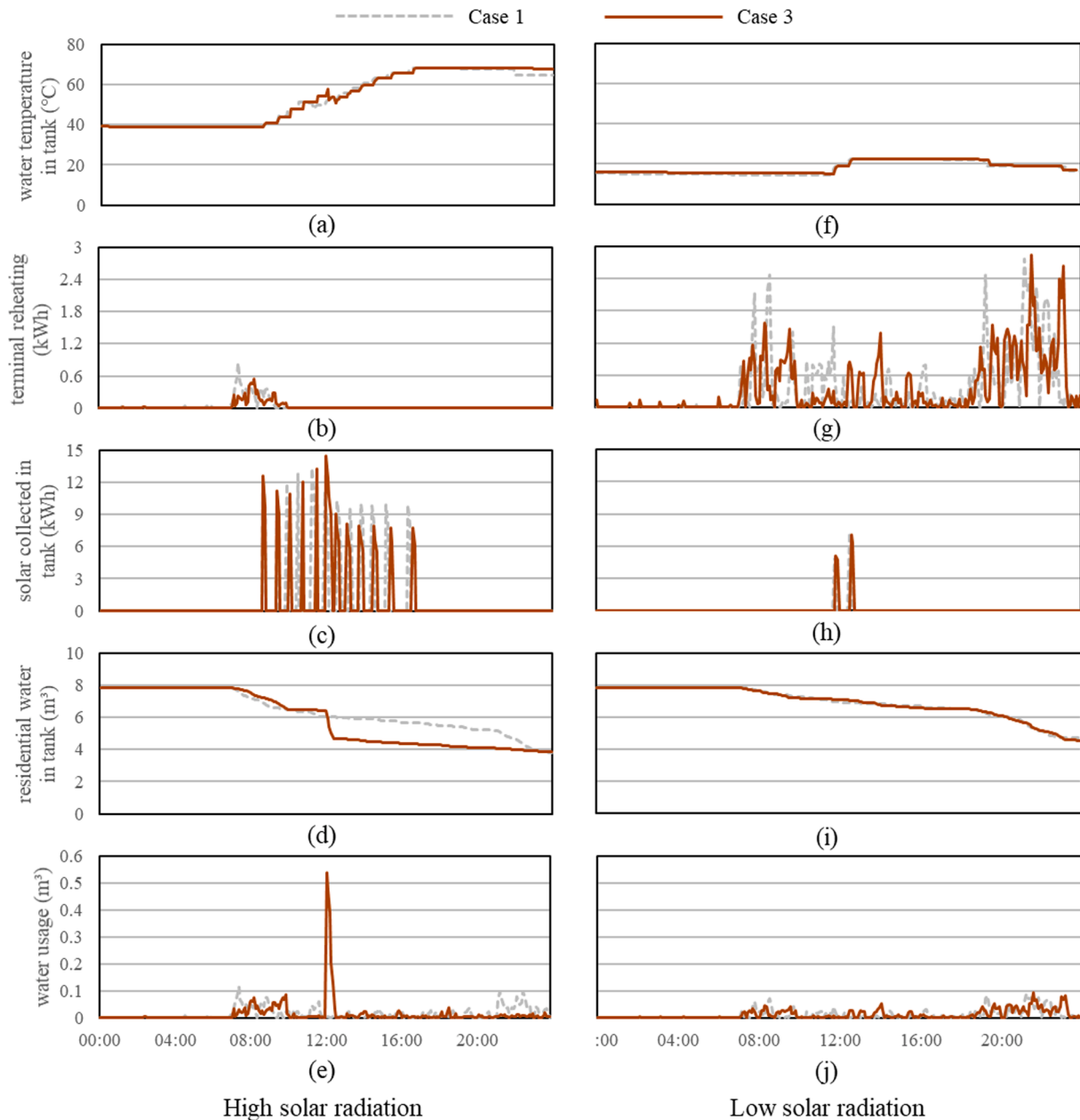
Fig. 11. Influence of water temperature feedback on the operation of standard designed CCCR system.

system, as indicated by the comparison of Cases 1 and 3. The Q_{r_CCCR} (high solar radiation: 17.84 kWh/m³, low solar radiation: 53.11 kWh/m³) and Q_{r_CCTR} (high solar radiation: 2.17 kWh/m³, low solar radiation: 31.55 kWh/m³) in case 3 were similar to that in case 1, which also proved the aforementioned conclusions.

The water temperature feedback achieved an energy-saving effect in the CCCR system through the optimized water tank capacity and pump control strategies, as shown in Case 4. During the days with high solar radiation, for a water tank of low capacity, as shown in Fig. 13 (a), the water temperature rose to 55 °C at 10:30, leading to a peak in hot water usage with the water temperature feedback in Case 4. The water amount in the tank was insufficient to match the excessive concentration of hot water demand, which led to the operation of the supply pump, and the water temperature in the tank dropped to 32.49 °C. The reheater

generated insufficient energy (7.60 kWh), as shown in Fig. 13 (b), and the collecting pump raised the water temperature in the tank to 49.19 °C. After the water usage peak, the water amount in the tank was approximately 1.74 m³, as shown in Fig. 13 (d). During the rest of the day, the water temperature in the tank was above 50 °C. This process showed that using water temperature feedback in Case 4 could move hot water usage in the evening to daytimes when solar radiation was abundant. Accordingly, the CCCR system could use solar energy to partially replace the electricity for heating the water, which further improves the solar energy utilization.

Meanwhile, owing to the low water usage at night, more hot water was left in the tank before the water was replenished early the next day. Furthermore, at the end of the day, the water temperature in the tank was 60 °C in both cases, while the remaining water in the tank was 2.77



Note:

Case 1: standard designed CCTR system;

Case 3: standard designed CCTR system with feedbacking water temperature in tank to users.

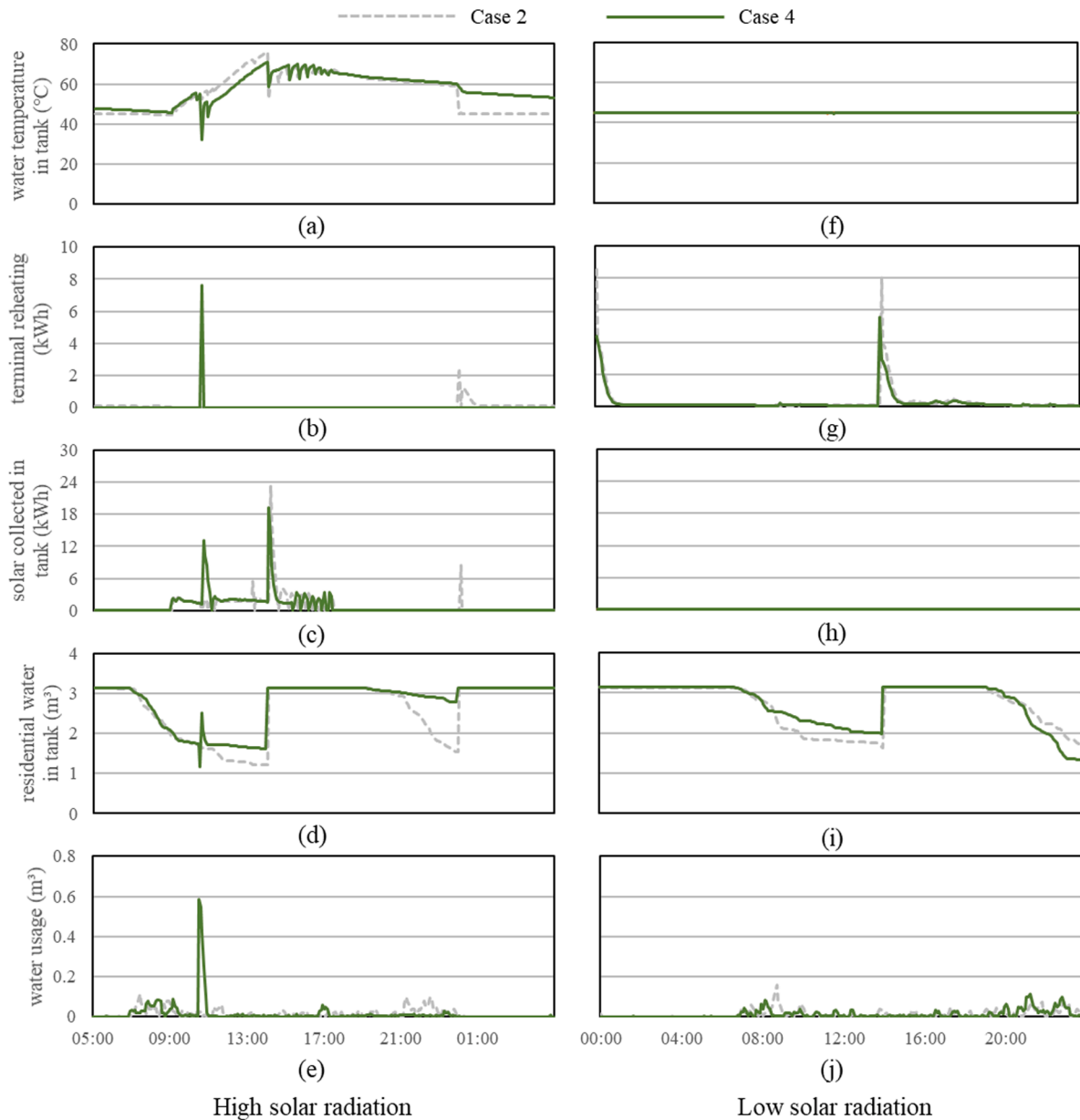
Fig. 12. Influence of water temperature feedback on the operation of standard designed CCTR system.

m^3 in Case 4, which was larger than that in Case 2 (1.54 m^3). Therefore, after the water was replenished at midnight, the water temperature in the tank was 45°C in Case 2 and 55°C in Case 4 (Fig. 13 (a)). Accordingly, no reheating energy was required in Case 4 for the second day from 0:00 to 9:30, whereas in Case 2, the reheater needed to provide 16.40 kWh of heat before sunrise.

After introducing the temperature signal to adjust the occupants' water usage profiles, based on the advantages of the aforementioned two periods, the solar energy utilization of CCTR during the days with high solar radiation could be further improved. For a typical day with strong solar radiation, the reheating energy consumption in Case 4 was 15.56 kWh , which was 31.49% of that in Case 2. The solar fraction that day was 88.37% in Case 4, which was 12.62% higher than that of Case 2. When the solar radiation was low, the water temperature in the tank

could not reach 55°C , but the occupants used hot water as usual; the reheating energy in Case 4 was close to that in Case 2.

Similar to the CCTR system, on a typical day with high solar radiation, the water temperature feedback strategy saved the reheating energy consumption from 0:00 to 9:30 in the CCTR system for Case 4, as shown in Fig. 14 (b). However, when the water temperature reached 55°C at 10:30, as shown in Fig. 14 (a), for Case 4, an excessive concentration of water usage triggered the supply pump. The water temperature in the tank dropped to 34.11°C , resulting in a heat supplement peak (6.24 kWh), as shown in Fig. 14 (b). It should be noted that, for the CCTR system, in the early morning, the reheater only needed to heat the water consumption on the user side, and the amount of hot water in this part was relatively small. Therefore, the electricity consumption of reheating that could be replaced by solar energy by adjusting the water



Note:

Case 2: optimized CCCR system based on actual DHW usage profile;

Case 4: optimized CCCR system based on actual DHW usage profile with feedbacking water temperature in tank to users.

Fig. 13. Influence of water temperature feedback on the operation of optimized CCCR system.

usage profile was relatively limited. Meanwhile, because of the small capacity of the water tank, excessively concentrated water usage peaks caused a shortage of water supply in the water tank. Therefore, a large amount of cold water was required to fill the tank, which caused a peak in the heat of the terminal reheaters and reduced the utilization of solar energy. For a typical day with high solar radiation, the solar fraction of the CCTR system in Case 4 was 96.34%, which was only 2.51% higher than that of Case 2.

On a typical day with low solar radiation, the water temperature in the tank of the CCCR and CCTR systems was lower than 55 °C, and users did not adjust their hot water usage behavior, which led to a similar result for reheating energy (Case 2: 112.32 kWh, Case 4: 121.25 kWh) and solar fraction in Case 2 (16.80%) and Case 4 (16.85%).

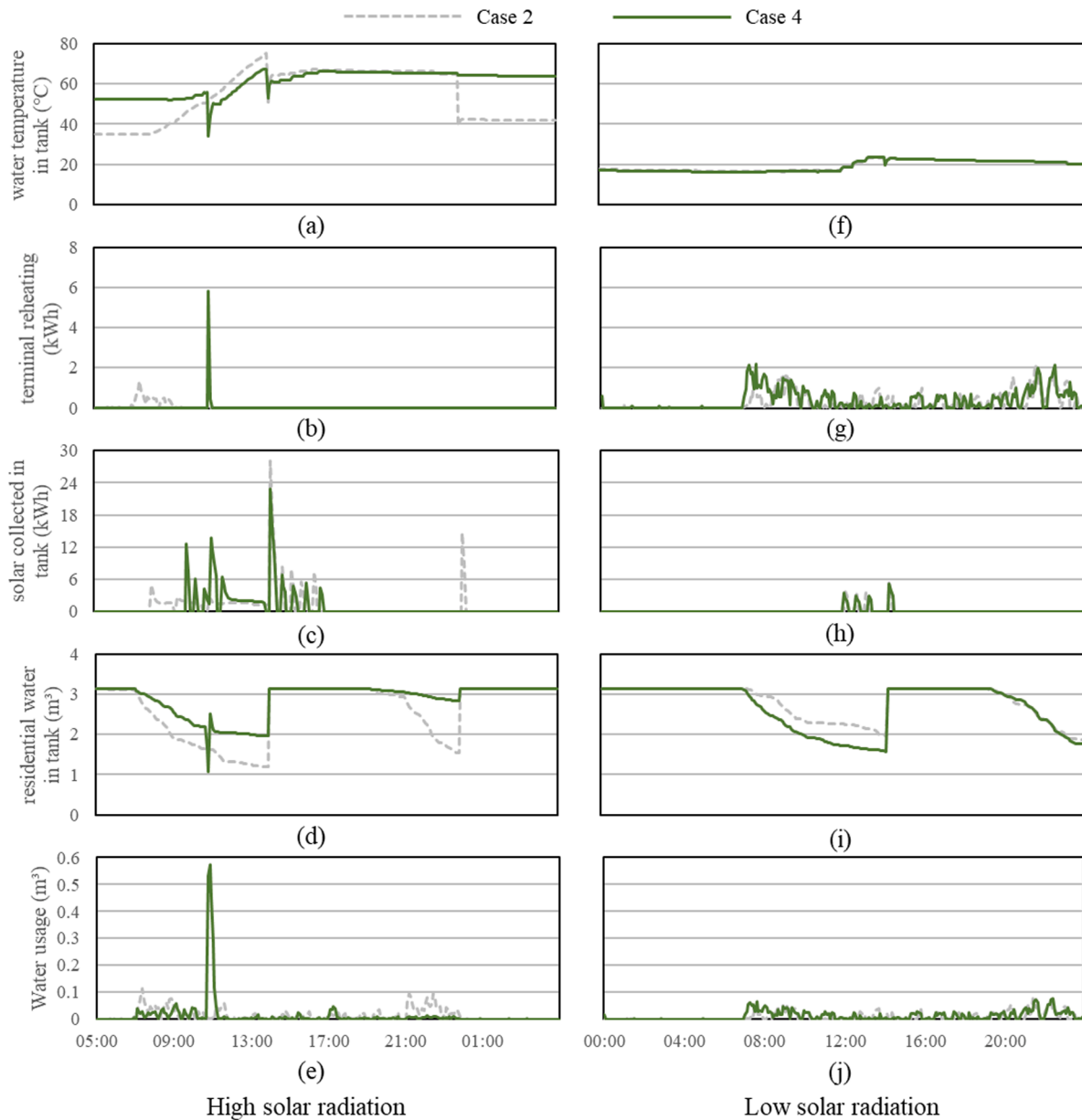
In Case 4, the Q_{F_CCCR} was 4.39 kWh/m³ on days with strong solar radiation, which was 34.54% of that in Case 2. On days with low solar

radiation, the Q_{F_CCCR} (50.78 kWh/m³) was similar to that in Case 2. For the CCTR system, the Q_{F_CCTR} was 2.10 kWh/m³ on high solar radiation days and 28.09 kWh/m³ on low solar radiation days, which was lower than that in Case 2.

Therefore, in Case 4, the feedback of water temperature to the user to adjust user behavior produced a significant energy-saving effect in the CCCR system compared to the CCTR system, which is consistent with f_{CCCR_A} and f_{CCTR_A} , as shown in Section 3.1.

4. Discussion

The analysis in Section 3 proved that providing information regarding water temperature in the tank to users to adjust their hot water usage behavior may improve the solar energy utilization of the SHW system. To further explore the influence of occupants' energy-



Note:

Case 2: optimized CCTR system based on actual DHW usage profile;

Case 4: optimized CCTR system based on actual DHW usage profile with feedbacking water temperature in tank to users.

Fig. 14. Influence of water temperature feedback on the operation of optimized CCTR system.

saving awareness, based on Case 4, f_{CCCR_A} and f_{CCTR_A} under different ratios of responding users and response probabilities were calculated, as shown in Figs. 15 and 16.

Previous studies have found that some users in SHW systems would adjust their water usage according to the water temperature in the DHW system [17]. Meanwhile, some studies proposed that among users with responsive behavior, their positivity is different [45]. Therefore, two parameters were considered in this study: the ratios of responding users and response probabilities. The ratio of responding users represented the proportion of responding users among all users, with a larger value indicating that more users in the system would take part in adjusting their water usage based on system information. The response probabilities represented the positivity of the responding users; the higher the value, the more likely users are to adjust their behavior when the feedback signal meets the requirements. The combination of the two

parameters indicated that the final number and intensity of occupants responded to the system information in the DHW system.

The annual solar fraction of the two SHW systems under different ratios of responding users and response probabilities showed similar trends. With the increase in the proportion of responding users and the response probability of the user, the overall f_{CCCR_A} and f_{CCTR_A} increased. This result proved that providing users with SHW system hot water temperature information, coupled with system optimization design, can effectively improve the utilization efficiency of solar energy. When 90% of users responded to the water temperature information, and the response probability was 70%, the CCTR system achieved a maximum f_{CCCR_A} of 44.3%, which was 8.5% higher than that of the system without information feedback. Based on the method mentioned in Section 3.1, in one household, 120.55 kWh of electricity can be saved in a year, and the annual CO₂ emission reduction would be 107.55 kg.

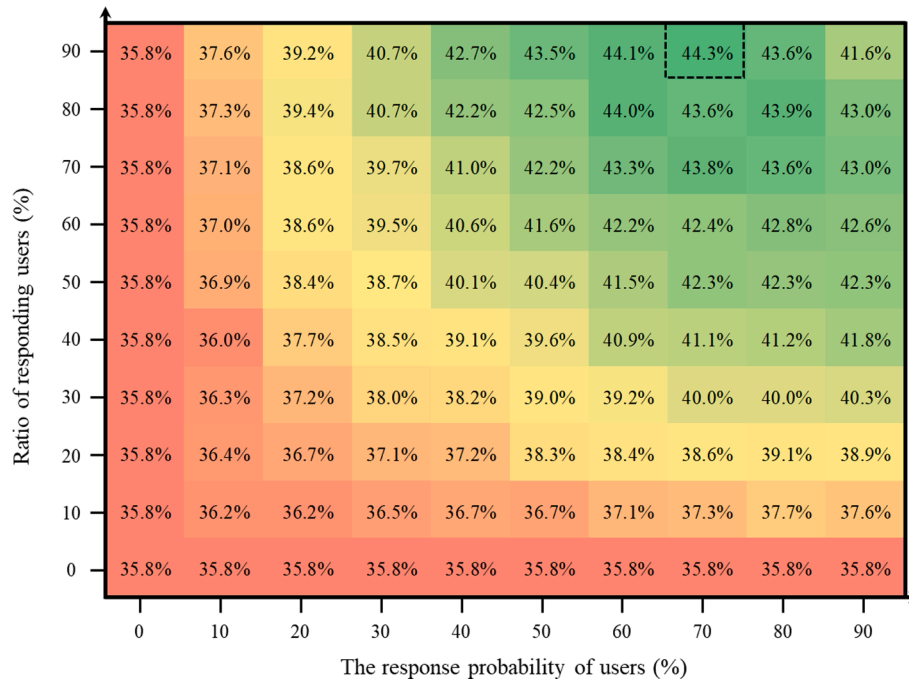


Fig. 15. Variations in solar fraction of optimized CCCR system with different ratios of responding users and response probability.

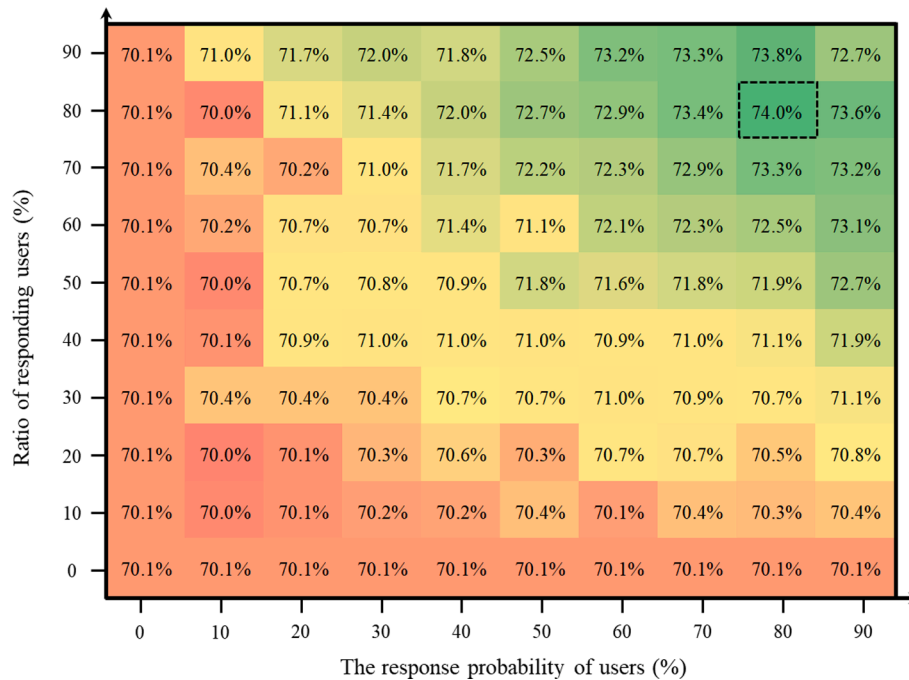


Fig. 16. Variations in solar fraction of optimized CCTR system with different ratios of responding users and response probability.

The cost savings would be 59.91 RMB per household per year.

For the CCTR system, there was also a specific combination of the ratio of responding users (80%) and response probability (80%) to achieve the maximum f_{CCTR-A} (74.0%), which was 3.9% higher than that of the system without information feedback.

Interestingly, unlike the CCCR system, in the CCTR system, providing water temperature to users does not always lead to higher energy efficiency even with an optimized system design, as shown in Fig. 16. When the ratio of responding users and response probability were 20% and 10%, respectively, the f_{CCTR-A} was close to that of the system without information feedback. This phenomenon further proved that the CCTR

system was not as sensitive to changes in the occupants' water usage profile, as in the CCCR system.

It should also be noted that a higher efficiency of solar energy utilization was achieved when the occupants took faster actions than when more occupants responded to the water temperature signal, as shown in Figs. 15 and 16. Several users reacting to the temperature signal simultaneously would cause an excessive peak in water consumption in the SHW system, resulting in the momentary failure of the SHW system, and electric reheating would be consumed to supplement it.

Therefore, if some technologies or policy methods are used to adjust the occupants' hot water demand, the water supply capacity of the SHW

system can be matched with the user's hot water demand, which can improve solar energy utilization efficiency, save energy consumption, and reduce CO₂ emissions.

In addition, the DHW behavior adjustment strategy was more suitable for centralized SHW systems containing families with elderly people, children, or freelancers. Compared to office workers with a more regular work schedule, the water use behavior of these people has a greater degree of freedom. Further research is needed to reveal, besides the water temperature in the tank, what kind of feedback information can effectively influence users' hot water usage behavior, and finally achieve the purpose of energy conservation and emission reduction.

5. Conclusions

The DHW usage behavior has an important impact on the energy consumption of the solar hot water (SHW) system, and the operating performance of the SHW system likewise influences the user's behavior. To reflect this universal interaction and quantitatively analyze its impact on solar energy utilization, an information-driven stochastic domestic hot water (DHW) behavior model was established and interacted with the SHW system model in this study. Subsequently, an analysis was conducted using two types of SHW systems: centralized collective and centralized reheating (CCCR) systems and centralized collective and terminal reheating (CCTR). The main conclusions drawn are as follows:

- (1) The proposed information-driven stochastic DHW usage behavior model could effectively describe the DHW usage profile and indicate the interaction between the SHW system and occupants' DHW usage.
- (2) The installation location of the reheater had a significant influence on the solar energy utilization of SHW systems. Because the energy consumption of the reheater was more synchronized with the actual heat demand of the occupants, the solar energy utilization of the CCTR system was approximately 30% higher than that of the CCCR system.
- (3) Based on the actual DHW usage profile, the optimal design and control strategy could save investment and increase the solar energy utilization effect of the SHW system. In the CCTR system, the optimal method can reduce the water tank capacity by 60%, and the initial investment can be reduced without compromising the performance of the system. For the optimized CCCR system, by improving the concurrence between the reheating energy and

users' DHW usage profile, the solar fraction could be improved, which was 11.5% higher than that of the standard designed CCCR system. Therefore, 163.10 kWh of electricity, 145.52 kg of CO₂ emission and 81.06 RMB can be saved per household per year.

- (4) The energy-saving effect of providing water temperature signals to users varied with different SHW systems and was influenced by the system design. Owing to the surplus water tank capacity, for the standard designed SHW system, water temperature feedback signals cannot achieve an energy-saving effect. For optimized SHW systems with small water tank capacity, this method can further achieve 8.5% and 3.9% higher solar fractions of CCCR and CCTR systems, respectively.
- (5) While considering the influence of occupants' hot water behavior, it was also necessary to focus on the matching ability of the SHW system. The results showed that excessive users reacting to the temperature signal simultaneously would cause an excessive peak in water consumption in the SHW system, which would cause a 2.7% reduction in solar energy utilization.

CRediT authorship contribution statement

Xin Zhou: Conceptualization, Methodology. **Shuai Tian:** Data curation, Writing – original draft. **Jingjing An:** Writing – review & editing. **Da Yan:** Supervision. **Lun Zhang:** Supervision. **Junyan Yang:** Supervision, Visualization, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Questionnaire for domestic hot water usage behavior survey

Dear Resident,

This survey focuses on the residents' usage behavior of domestic hot water and energy conservation. We hope to analyze the current situation of domestic hot water usage and energy consumption by Chinese residents through your feedback information. Your information is valuable to us.

This survey is anonymous. We guarantee complete anonymity when using your information and will not disclose your personal thoughts to any other organization or individual. This study does not pose any risks to you.

Thank you for participating in this survey.

Part 1: The basic information of users.

1. How old are you?

Under 20 years old	20–29 years old	30–39 years old	40–49 years old	50–59 years old	Over 60 years old
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2. Your gender is?

Male	Female
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3. Your current city of residence is?

Province:_____.	City:_____.
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Part 2: Domestic hot water usage behavior survey.

1. How do you use your domestic hot water ?

Shower	Washing face	Kitchen work	Washing clothes	Cleaning the house	Others
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2. How many times do you shower in a week?

Spring and autumn	>7	7	3–4	1–2
Summer	>7	7	3–4	1–2
Winter	>7	7	3–4	1–2

3. How long do you spend showering?

Spring and autumn	5–10 min	10–20 min	20–30 min	>30 min
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Summer	5–10 min	10–20 min	20–30 min	>30 min
Winter	5–10 min	10–20 min	20–30 min	>30 min

4. When do you shower every day?

Spring and autumn	5:00–9:00	9:00–12:00	12:00–16:00	16:00–19:00	19:00–22:00	22:00–24:00	24:00–5:00
Summer	5:00–9:00	9:00–12:00	12:00–16:00	16:00–19:00	19:00–22:00	22:00–24:00	24:00–5:00
Winter	5:00–9:00	9:00–12:00	12:00–16:00	16:00–19:00	19:00–22:00	22:00–24:00	24:00–5:00

5. What is the temperature of your hot water when you take a shower?

Spring and autumn: _____ °C	Summer: _____ °C	Winter: _____ °C
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6. Do you take a shower or bath?

Shower	Bath	Both	Other: _____.
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7. **If you usually take a shower**, what is the percentage of duration of shower head open over the total shower?

<50%	50%–70%	70%–90%	100% (always on)
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Part 3: Building and type of domestic hot water system

1. How many floors are there in your residence?

1–3 floors	4–6 floors	7–9 floors	Over 10 floors
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2. Is the domestic hot water centrally prepared and supplied in your house?

Yes, provided by the community/building uniformly	No, using household DHW water heater
No, using: _____ water heater	No, no water heater

3. What is the type of heat source for domestic hot water supply in your house?

Natural gas	Electricity	Solar energy	Air source	Other: _____.
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If the domestic hot water in your house is provided by community/building uniformly, please answer questions 4

to 6.

4. Do you know the monthly usage amount of domestic hot water in your house?

Yes, it is probably _____ ton/month	No, I don't know
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5. Do you know the monthly cost of domestic hot water in your house?

<10 RMB/t	10–15 RMB/t	15–20 RMB/t	20–25 RMB/t	Over 25 RMB/t	Don't know
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6. Do you think the cost of domestic hot water is expensive?

Yes, it is expensive	It is a little expensive	No, it is an acceptable price	No, it is cheap
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If you have a solar hot water system in your home, please answer questions 7–9.

7. What type of the solar hot water system do you have in your house?

Centralized collective and centralized reheating system	Centralized collective and terminal reheating system
Household solar water heater (collector is balcony wall-mounted)	Household solar water heater (collector is roof-mounted)

8. What energy is used for auxiliary heating in your solar hot water system?

Natural gas	Electricity	Ground source heat pump	Other: _____.	Don't know
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9. What improvements would you recommend for solar hot water system?

Thank you again for participating in this survey!

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